Page

seq $\{S_n\}$

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots$$

seq $\{S_n\}$

The sequence S_n is called sequence of partial sum.

The series $\sum a_n$ is said to be convertion if the sequence of partial sum $\{S_n\}$

i.e. ① If $\lim_{n\to\infty} S_n$ = finite then $\sum a_n$ is Convergent

~~Divergent~~

② If $\lim_{n\to\infty} S_n = \pm\infty$, then $\sum a_n$ is divergent

③ If $\lim_{n\to\infty} S_n$ is not Unique as $n \to \infty$ then $\sum a_n$ is oscillation.

Note: The nature of the series remains unchanged if finite number of terms are added or omitted from the series.

Theorem

If $\sum_{n=1}^{\infty} u_n$ is Convergent

Then $\lim_{n\to\infty} u_n = 0$ $[p \to q]$

Note: but the converse ~~of~~ this is not true

eg. $\sum 1/n$ $(1 + 1/2 + 1/3 \cdots)$

If $\lim \sum_{n=1}^{\infty} u_n \neq 0$ $p \Leftrightarrow q$.

then, divergent $\sum u_n$

1) $\sum_{n=1} \cos(1/n)$

$u_n = \lim_{n\to\infty} \cos(1/n)$

$= \cos(0)$

$\boxed{1}$